

Guidance document on broth microdilution testing of cefiderocol

January 2024

Introduction

Cefiderocol is a siderophore cephalosporin with activity against aerobic Gram-negative bacteria. It is approved for the treatment of aerobic Gram-negative infections in adults with limited therapeutic options. EUCAST has determined clinical breakpoints (MIC and zone diameter) for *Enterobacterales* and *Pseudomonas aeruginosa*. Cefiderocol exhibits pronounced activity against *Acinetobacter baumannii* and *Stenotrophomonas maltophilia* but there was insufficient clinical evidence to determine breakpoints.

Cefiderocol shows activity against many isolates with high MIC values for and resistance to other beta-lactam agents, both cephalosporins and carbapenems. However, resistance to cefiderocol exists and these organisms can be detected with antimicrobial susceptibility testing (AST) performed with MIC or disk diffusion AST.

Disk diffusion is performed with a cefiderocol 30 µg disk according to EUCAST standard recommendations for non-fastidious organisms, see [EUCAST Disk Diffusion Manual](#).

Broth microdilution testing methodology for cefiderocol

Cefiderocol requires low iron levels for optimal activity. Broth microdilution MIC determination must be performed in iron-depleted Mueller-Hinton broth. The iron concentration of liquid medium affects reproducibility and therefore should be strictly defined. Chelation is used for the iron depletion, and this procedure also removes other cations (i.e. calcium, magnesium and zinc). Following this process, cations are added back to concentrations of calcium 20–25 mg/L, magnesium 10–12.5 mg/L, and zinc 0.5–1.0 mg/L. The procedure necessary to produce the broth is described in **Appendix A**.

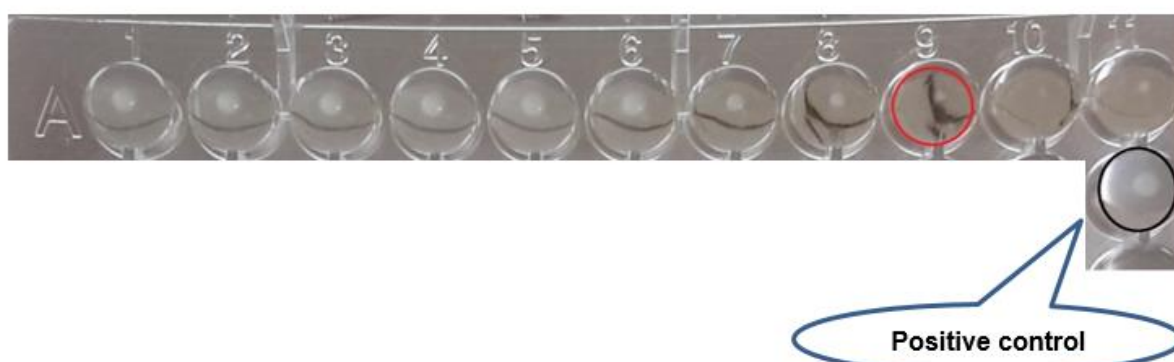
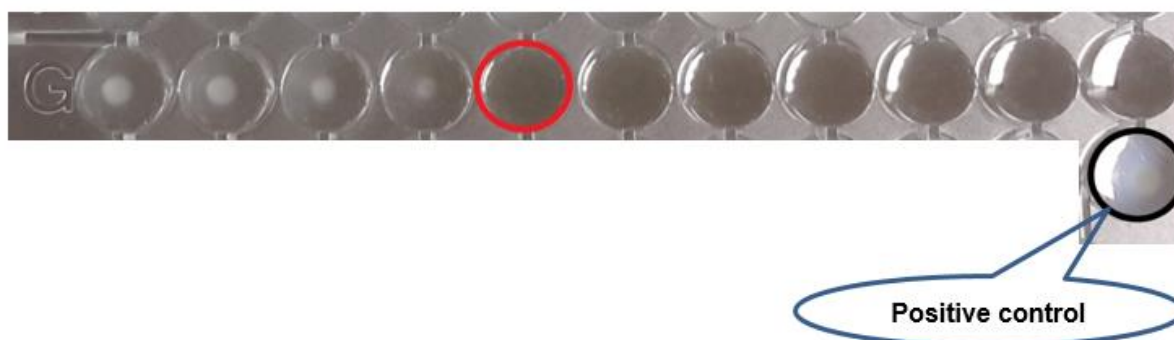
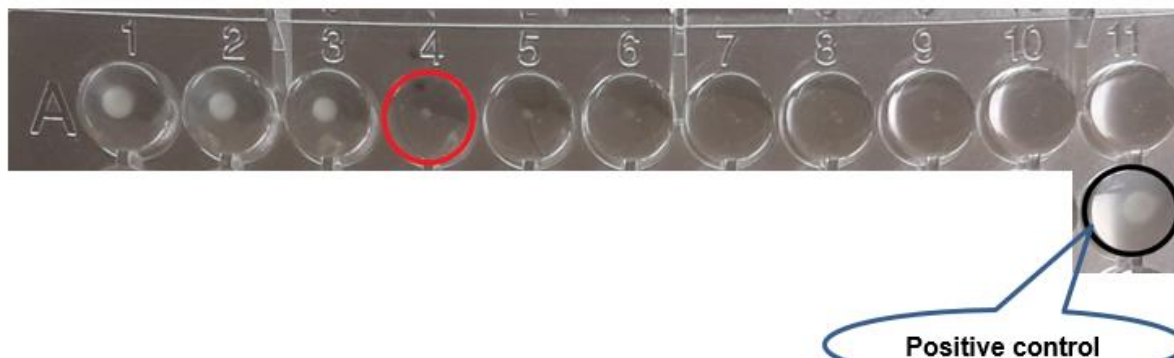
Reading of cefiderocol broth microdilution MIC endpoints

- The MIC is read as the first well in which the reduction of growth corresponds to a button of <1 mm or is replaced by the presence of light haze/faint turbidity (≥80% reduction in haze as compared to the growth control).
- The positive control should show strong growth in the form of a button of >2 mm or heavy turbidity.

See pictures with reading examples on the next page.

Examples of reading cefiderocol MIC endpoints

The MIC of cefiderocol is read as the first well in which a reduction of growth corresponding to a button of <1 mm or the presence of light haze/faint turbidity is observed (red ring). The positive control should show strong growth (button of >2 mm or heavy turbidity). Cefiderocol concentrations increase two-fold serially from column 1 (X mg/L) to column 11 (Y mg/L) wells.



Appendix A

Preparation of iron-depleted cation-adjusted Mueller Hinton Medium (ID-CAMHB)¹

1. Add 100 g Chelex100® (Bio-Rad) to 1 L of autoclaved Mueller Hinton broth (other cation-binding resin can be used in place of Chelex)
2. Stir at room temperature for approximately 6 hours
3. Filter using 0.2 micron filter (The cation concentration of Mg, Ca, Zn and Fe should be below detection limit by these treatments.)
4. Add calcium (final concentration 20-25 µg/mL), magnesium (final concentration 10-12.5 µg/mL) and zinc (final concentration 0.5-1.0 µg/mL) into the Chelex-treated medium
5. Adjust pH to 7.2-7.4 by using HCl
6. Filter using 0.2 micron filter (optional)

Defined cation concentration in ID-CAMHB

Ca	20-25 µg/mL
Mg	10-12.5 µg/mL
Zn	0.5-1.0 µg/mL
Fe	0.03 µg/mL or less

The iron concentration can be detected by using VISOCOLOR® HE Iron (Machere Nagel, Germany) with detection limit of 0.01 µg/mL.

¹ Based on the procedure described by Hackett et al [1] with the modification to stir with Chelex for 6 hours instead of 2 hours.

References

1. Hackel MA, Tsuji M, Yamano Y, Echols R, Karlowsky JA, Sahm DF. *Reproducibility of broth microdilution MICs for the novel siderophore cephalosporin, cefiderocol, determined using iron-depleted cation-adjusted Mueller-Hinton broth.* Diagn Microbiol Infect Dis. 2019 Aug;94(4):321-325.